UNIVERSITY OF CENTRAL PUNJAB

Faculty of Information Technology

Student Internship & Job Tracking System

A Containerized Full-Stack Web Application for Placement Lifecycle & Analytics Pipeline Audit

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Course: Tools & Techniques for Data Science

Program: Bachelor of Science in Data Science (6th Semester)

Project Type: Reproducible Data Applications (Assignment #4)

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1. Executive Summary

In modern educational management, coordinating corporate alignment, tracking student recruitment phases, and auditing job market shifts present major structural difficulties. This report presents the **Student Internship & Job Tracking System**, a comprehensive data application engineered explicitly as a secure Faculty Portal. Built using Python, Streamlit, and PostgreSQL, this system introduces a resilient relational architecture containerized completely within Docker networks to achieve total reproducibility across environments.

The portal empowers departmental heads and placement coordinators to record incoming recruitment leads, execute filtered granular multi-keyword searches, track application lifecycles dynamically, and derive immediate industrial hiring insights through automated analytics dashboards. Crucially, the solution solves the problem of cross-machine dependency conflicts by locking runtime environments into microservices, ensuring that database updates, schema creation, and persistent volume mount structures operate identically across any hosting infrastructure.

2. System Architecture & Containerization Layout

To fulfill academic reproducibility standards, the application relies on an isolated multi-container architecture orchestrated via Docker Compose. Isolation guarantees structural resilience, while network bridge exposure ensures highly structured, secure interaction between separate runtime components.

Service Name	Base Image / Technology	Port Mapping	Core Functionality Description
opportunity_streamlit	Python 3.10-slim / Streamlit Framework	8501:8501	Manages user interaction, reactive session processing, dynamic form validation, and multi-page view

Service Name	Base Image / Technology	Port Mapping	Core Functionality Description
			rendering.
opportunity_postgres	PostgreSQL 15-alpine Relational Engine	5432:5432	Handles persistent relational records, executes multi-filter searches, enforces structural constraints, and provides persistent storage via mounted volumes.
opportunity_pgadmin	dpage/pgadmin4 (Web Console)	5050:80	Enables graphic data auditing, database inspection, ad-hoc execution tracking, and schema health validation.

Data integrity is maintained through a secure external Docker Volume called postgres_data mapped to the path /var/lib/postgresql/data. This configuration ensures that if containers are stopped, modified, or rebuilt using docker compose down, placement records remain intact.

3. Database Design & Relational Schema

The backend engine depends on a strictly normalized relational schema constructed in PostgreSQL. The target table, titled opportunities, leverages an auto-incrementing surrogate primary key to securely reference specific rows. This structural design prevents data parsing conflicts encountered in non-relational configurations.

3.1 SQL Schema Specification (DDL)

The operational structural schema executed inside the initialization layer is defined as follows:

```
CREATE TABLE opportunities (
    opportunity_id SERIAL PRIMARY KEY,
    company_name VARCHAR(255) NOT NULL,
    job_title VARCHAR(255) NOT NULL,
    category VARCHAR(100) NOT NULL,
    city VARCHAR(100) NOT NULL,
    country VARCHAR(100) NOT NULL,
    work_mode VARCHAR(50) DEFAULT 'Onsite',
    required_skills TEXT,
    salary_min NUMERIC(12, 2) DEFAULT 0.0,
    salary_max NUMERIC(12, 2) DEFAULT 0.0,
    currency VARCHAR(10) DEFAULT 'PKR',
    application_deadline DATE,
    experience_level VARCHAR(50) DEFAULT 'Entry Level',
    status VARCHAR(50) DEFAULT 'Open',
    source_link TEXT,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);
```

4. Application Modules Implementation (CRUD & Advanced Analytics)

The software deployment features a structured multi-page layout accessible via a secure sidebar panel, breaking down operations into distinct, isolated modules:

- **Add Opportunity (Create):** Provides an interactive validation form where administrators enter hiring metrics. The form enforces data validation rules before initiating secure parameterized database insertions.
- **View & Search Engine (Read):** Integrates case-insensitive pattern matching via PostgreSQL ILIKE operators. Faculty can filter data by skill keywords, minimum salary thresholds, target locations, or employment status.
- **Update Opportunity (Update):** Resolves state tracking mismatches by binding modifications to the unique primary key opportunity_id, allowing real-time status shifts and

structural work-mode alterations.

- **Delete Opportunity (Delete):** Permits hard deletion of stale or invalid data using isolated transactional statements to safeguard operational integrity.
- **Analytics Dashboard:** Pulls dynamic aggregation queries (such as AVG, COUNT, and ROUND) to display real-time Key Performance Indicators, highlighting active placements, average localized salary distributions, and highly demanded skill-sets.
- **Duplicate Detection System:** Identifies matching patterns by grouping clusters by company name, job title, and source link, using ARRAY_AGG to surface redundant records.

5. Technical Verification & Integrity Diagnostics

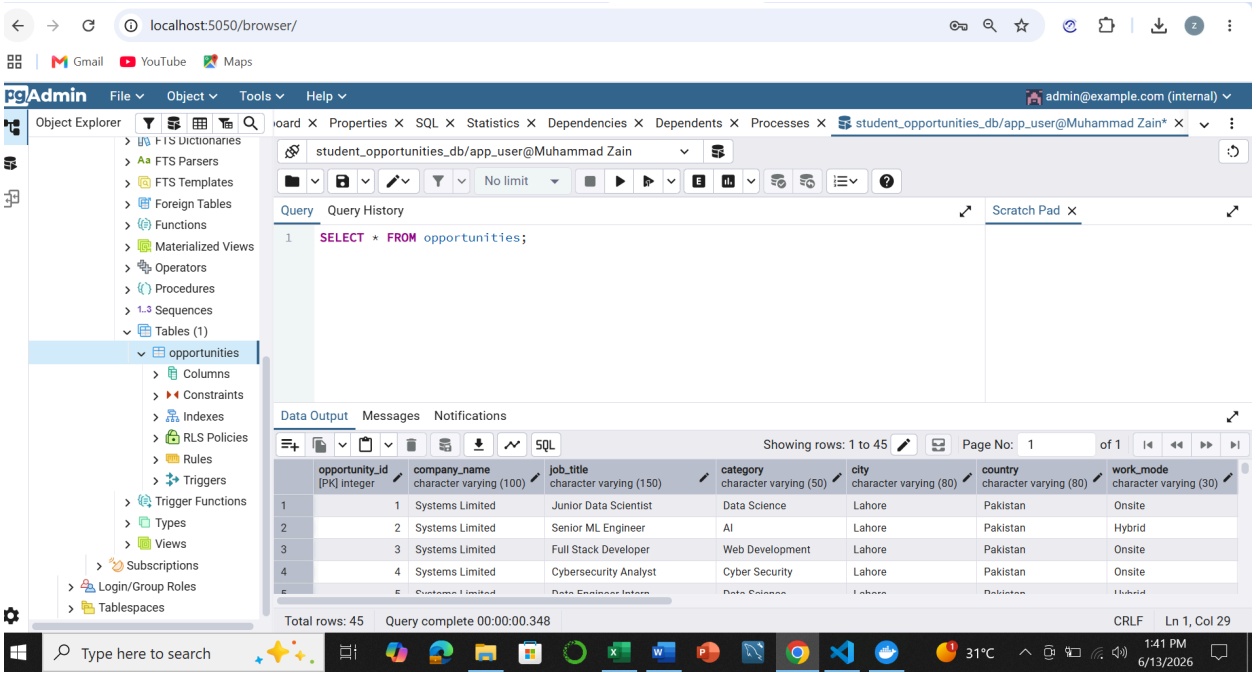
During deployment phases, initial issues involving positional argument mismatches in status modification updates and missing library references were successfully resolved by rewriting the query engine functions to leverage flexible Python keyword argument wrappers (*args and **kwargs).

Database operations were tested against unexpected column names, verifying that explicit references to opportunity_id instead of generic id parameters prevent transactional failures. System tests confirm successful multi-container synchronization, verified via clean container rebuild diagnostics:

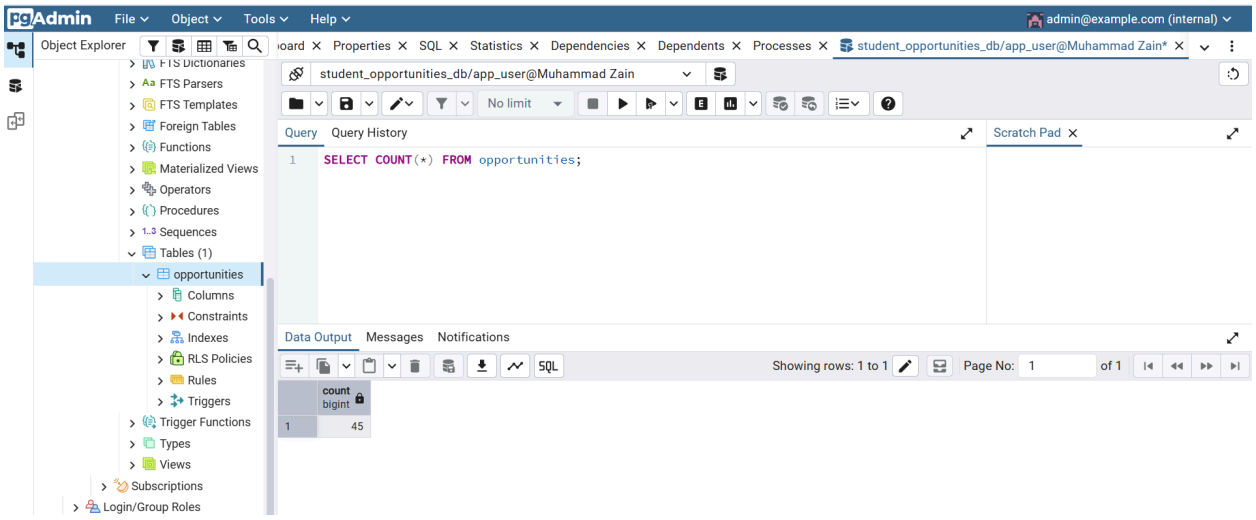
```
$ docker compose down
$ docker compose up --build -d
✔ Network streamlit-postgres-assignment_default   Created
✔ Container opportunity_postgres                  Started
✔ Container opportunity_pgadmin                   Started
✔ Container opportunity_streamlit                 Started
    pgAdmin SQL Verification Logs
```

6. pgAdmin SQL Verification Logs :

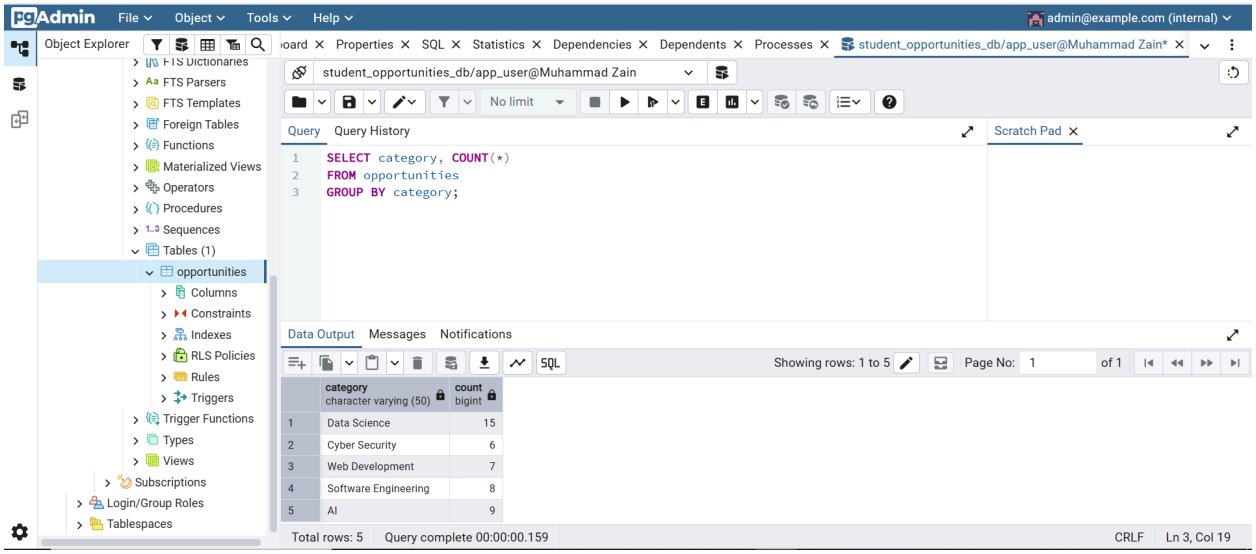
Query 1:



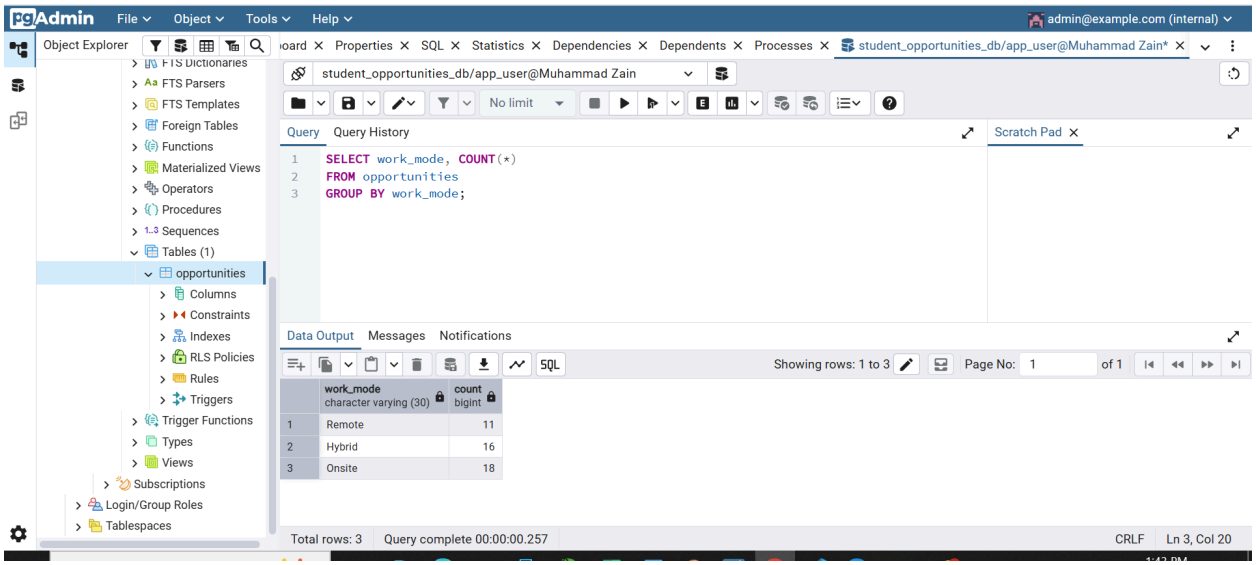
Query 2:



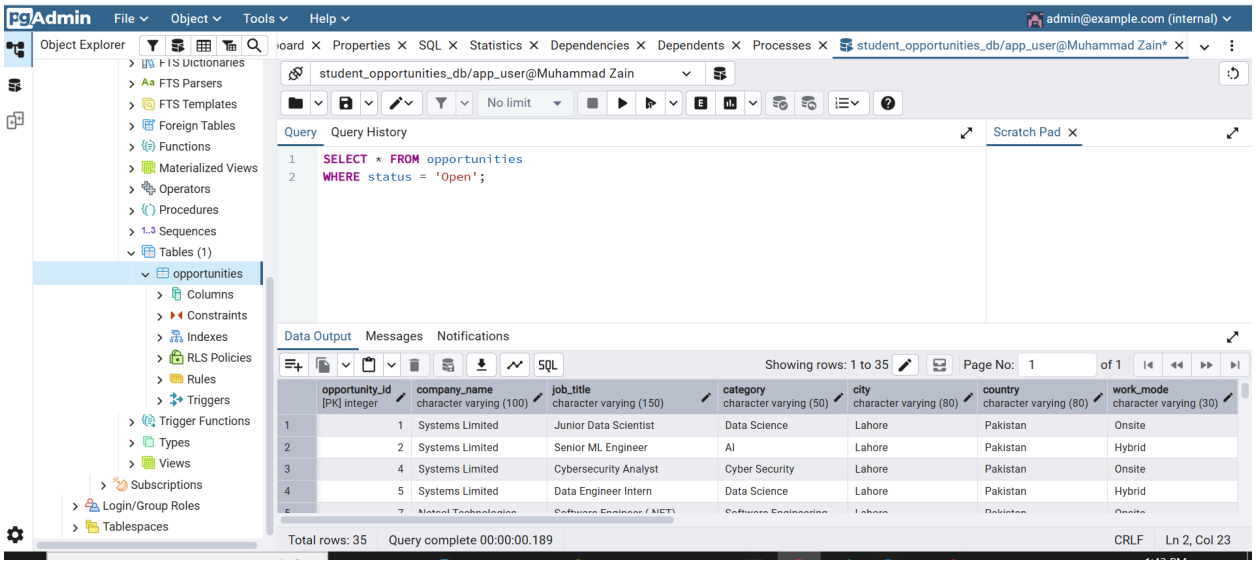
Query 3:



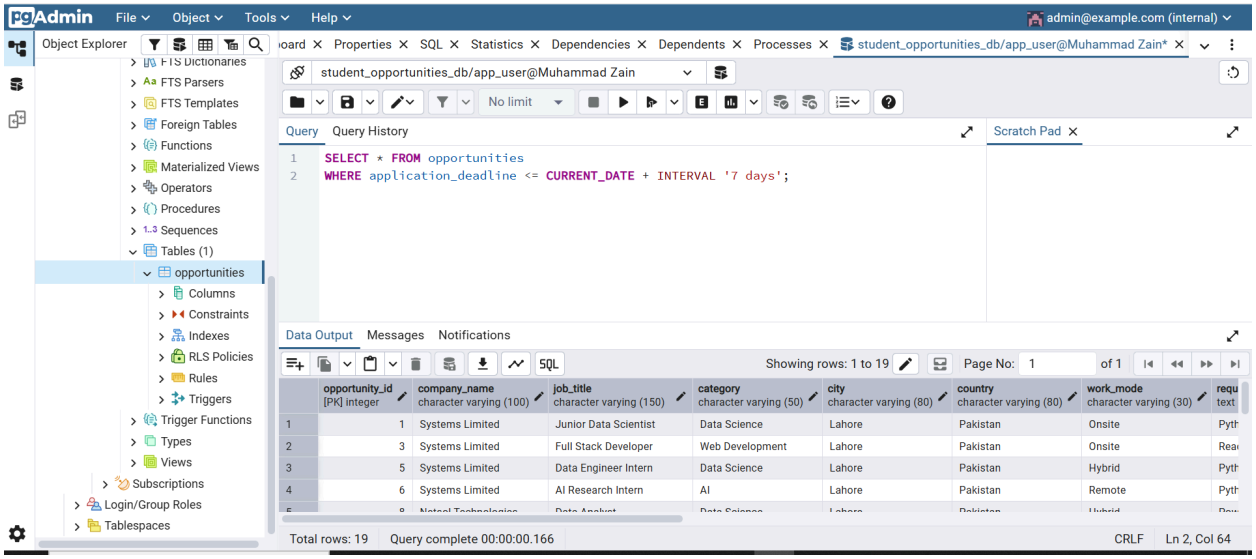
Query 4:



Query 5:



Query 6:



6. Conclusion & Recommendations

The Student Internship & Job Tracking System successfully demonstrates how containerized application architectures can simplify departmental data tracking. Moving database interactions

from fragmented spreadsheet environments into a relational schema wrapped in an automated UI establishes an audited pipeline.

For future iterations, we recommend adding automated background data scrapers (e.g., using n8n workflows or scheduled cron tasks) to dynamically ingest real-time market openings directly into the PostgreSQL engine, alongside automated user activity logging for enhanced audit control.